

Agentic AI Project

- Agentic Tool-Augmented LLMs for Indic Voice-Based Task Automation
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Problem Statement

- LLMs are limited to text and cannot execute real-world tasks.
- Lack of voice support for Indian users.

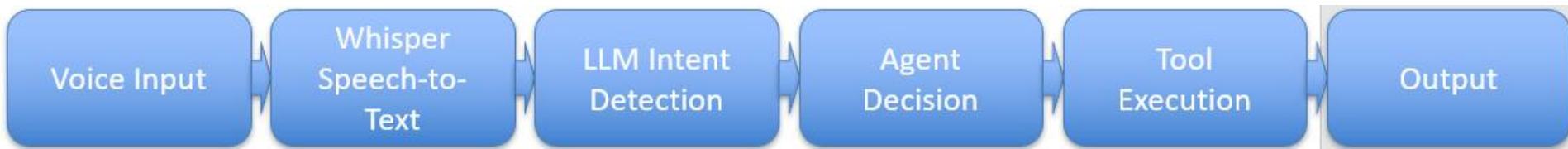
Proposed Solution

- Agentic AI system that:
 - - Takes voice input
 - - Understands intent
 - - Selects tools
 - - Executes tasks

System Architecture

- Voice → Whisper → LLM → Agent → Tool → Output

Visual Architecture Diagram



Technologies Used

- Python
- Whisper
- Transformers
- Datetime
- Google Colab

Implementation

- Voice + Text input
- Intent detection
- Tool execution (time, calculator)

Results

- Text: '5+10' → 15
- Voice: 'Abhi time kya hai' → current time
- Accurate intent detection

Demo

- 1. Text input demo

4. Quick text demo

This is live demo.

```
[18] ✓ 0s
▶ tests = [
    "what is the time now?",
    "calculate 5 + 10 * 2",
    "Namaste",
    "time kya hai",
    "5 plus 7"
]

for t in tests:
    print(run_agent(t))

... {'input': 'what is the time now?', 'intent': 'get_time', 'output': '2026-03-30 09:20:31 AM IST'}
{'input': 'calculate 5 + 10 * 2', 'intent': 'calculator', 'output': '25'}
{'input': 'Namaste', 'intent': 'greeting', 'output': 'Namaste! I am your agentic AI assistant.'}
{'input': 'time kya hai', 'intent': 'get_time', 'output': '2026-03-30 09:20:31 AM IST'}
{'input': '5 plus 7', 'intent': 'calculator', 'output': '12'}
```



[Open Live Colab Demo](#)

2. Voice input demo using Whisper

5. Voice input with Whisper

Upload an audio file and transcribe it.

You can upload `.wav`, `.mp3`, `.m4a`, etc.

 [Open Live Colab Demo](#)

```
[4] ✓ 2m
from google.colab import files
uploaded = files.upload()

audio_file = list(uploaded.keys())[0]
print("Uploaded file:", audio_file)

... Choose Files Recording.m4a
Recording.m4a(audio/x-m4a) - 98313 bytes, last modified: 3/30/2026 - 100% done
Saving Recording.m4a to Recording.m4a
Uploaded file: Recording.m4a

[7] 14s
import whisper

model = whisper.load_model("base")
result = model.transcribe(audio_file)

transcribed_text = result["text"]
print("Transcribed text:", transcribed_text)

... 100%|████████████████████████████████████████| 139M/139M [00:01<00:00, 133MiB/s]
/usr/local/lib/python3.12/dist-packages/whisper/transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")
Transcribed text: Hello, who are you?

[8] 0s
agent_result = run_agent(transcribed_text)
print(agent_result)

{'input': ' Hello, who are you?', 'intent': 'greeting', 'output': 'Namaste! I am your agentic AI assistant.'}
```

Challenges

- Audio processing issues
- Simple intent detection
- Limited tools

Future Scope

- Add APIs (booking, maps)
- Multi-agent systems
- Full automation assistant

Conclusion

- Agentic AI enables autonomous task execution.
- Step towards real-world AI assistants.

Executive Summary

- • Built an agentic AI assistant for Indic voice commands
- • Uses Whisper for speech-to-text and LLM-driven intent routing
- • Demonstrates autonomous tool usage (time + calculator)
- • Supports both text and voice demos in Google Colab
- • Strong future scope: API automation, multi-agent workflows

Evaluation Alignment

- Problem Understanding → Accessibility gap in Indic voice AI
- Implementation → End-to-end working prototype in Colab
- Results → Correct tool selection and successful outputs
- Viva → Clear distinction between LLMs and agentic systems
- Documentation → Structured report with results + references

References

- 1. OpenAI Whisper: Robust Speech Recognition via Large-Scale Weak Supervision (2022)
- 2. Hugging Face Transformers Documentation
- 3. ReAct: Synergizing Reasoning and Acting in Language Models (Yao et al., 2023)
- 4. LangChain Agents and Tool Calling Documentation
- 5. Tool-Augmented Language Models for Autonomous Task Execution
- 6. Review 1 project concept and methodology baseline from submitted PDF

Thank You

Questions & Viva